



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

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Computer Networks/0705210503

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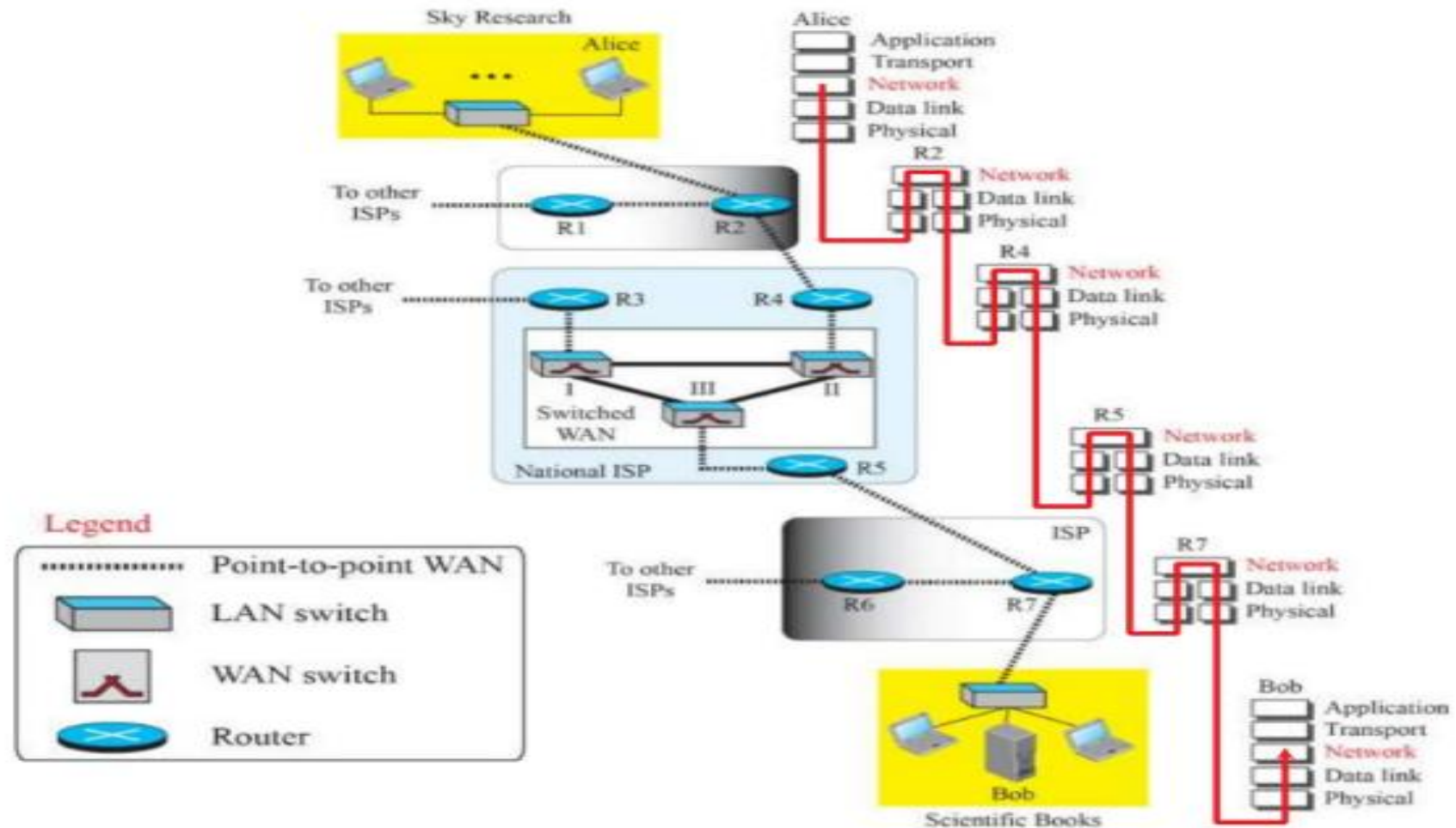
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Figure 18.1: Communication at the network layer





18.18.1 Packetizing

The first duty of the network layer is definitely packetizing: encapsulating the payload in a network-layer packet at the source and decapsulating the payload from the network-layer packet at the destination. In other words, one duty of the network layer is to carry a payload from the source to the destination without changing it or using it. The network layer is doing the service of a carrier such as the postal office, which is responsible for delivery of packages from a sender to a receiver without changing or using the contents.



Functions of Network Layer

- **Routing:** When a packet reaches the router's input link, the router will move the packets to the router's output link. For example, a packet from S1 to R1 must be forwarded to the next router on the path to S2.
- **Logical Addressing:** The data link layer implements the physical addressing and network layer implements the logical addressing. Logical addressing is also used to distinguish between source and destination system. The network layer adds a header to the packet which includes the logical addresses of both the sender and the receiver.
- **Internetworking:** This is the main role of the network layer that it provides the logical connection between different types of networks.
- **Fragmentation:** The fragmentation is a process of breaking the packets into the smallest individual data units that travel through different networks



Responsibilities of Network Layer

➤ Logical Addressing

- In the internet world, there are two kinds of addressing implemented by the data link layer, it handles addressing problems locally.
- If the network passes through the network boundary, we need another system to distinguish source and destination systems.
- The logical addressing at the network layer while physical addressing at the data link layer is defined by the MAC address of a device, whereas the IP addressing is determined at the network layer of the OSI model. This addressing is also called as logical addressing.
- The network layer adds a header to the packet which is coming from the upper layer includes the logical addresses of the sender and receiver.

➤ Routing

- When two independent networks or links are attached to create an internetwork that is the network of networks or a large network, the connecting devices route the packets to its destination.
- The forwarding of the data request to servers is known as routing.



Responsibilities of Network Layer

Fragmentation and Reassembly

- The network layer must send data down to the data link layer for transmission. The data or information that the network layer receives is in the form of a packet and the data that data link layer forwards is called a frame
- The network layer has the responsibility of Fragmentation and reassembly because some data link layer technologies have limits on the length of any message that can be sent.
- If the packet of data that the network layer has to send is too large, the network layer must break the packet up, send each packet to the data link layer, and then have pieces reassembled once they arrive at the network layer on the destination system

Example:

- If I want to access some data from Facebook then I will open my laptop, type URL of Facebook and send an HTTP request to facebook.com for some data.
- Since the server of Facebook is situated outside my local area network, my request is forwarded to Facebook through the default gateway or router of my institution



18-2 PACKET SWITCHING

From the discussion of routing and forwarding in the previous section, we infer that a kind of switching occurs at the network layer. A router, in fact, is a switch that creates a connection between an input port and an output port (or a set of output ports), just as an electrical switch connects the input to the output to let electricity flow.



18.2.1 Datagram Approach

When the Internet started, to make it simple, the network layer was designed to provide a connectionless service in which the network-layer protocol treats each packet independently, with each packet having no relationship to any other packet. The idea was that the network layer is only responsible for delivery of packets from the source to the destination. In this approach, the packets in a message may or may not travel the same path to their destination. Figure 18.3 shows the idea..



Datagram

- This approach uses a different, more dynamic scheme, to determine the route through the network links.
- Each packet is treated as an independent entity, and its header contains full information about the destination of the packet.
- The intermediate nodes examine the header of the packet, and decide to which node to send the packet so that it will reach its destination.
- In this method, the packets don't follow a pre-established route, and the intermediate nodes (the routers) don't have pre-defined knowledge of the routes that the packets should be passed through.
- Packets can follow different routes to the destination, and delivery is not guaranteed .
- Due to the nature of this method, the packets can reach the destination in a different order than they were sent, thus they must be sorted at the destination to form the original message.
- This approach is time consuming since every router has to decide where to send each packet.
- The main implementation of Datagram Switching network is the Internet, which uses the IP network protocol.

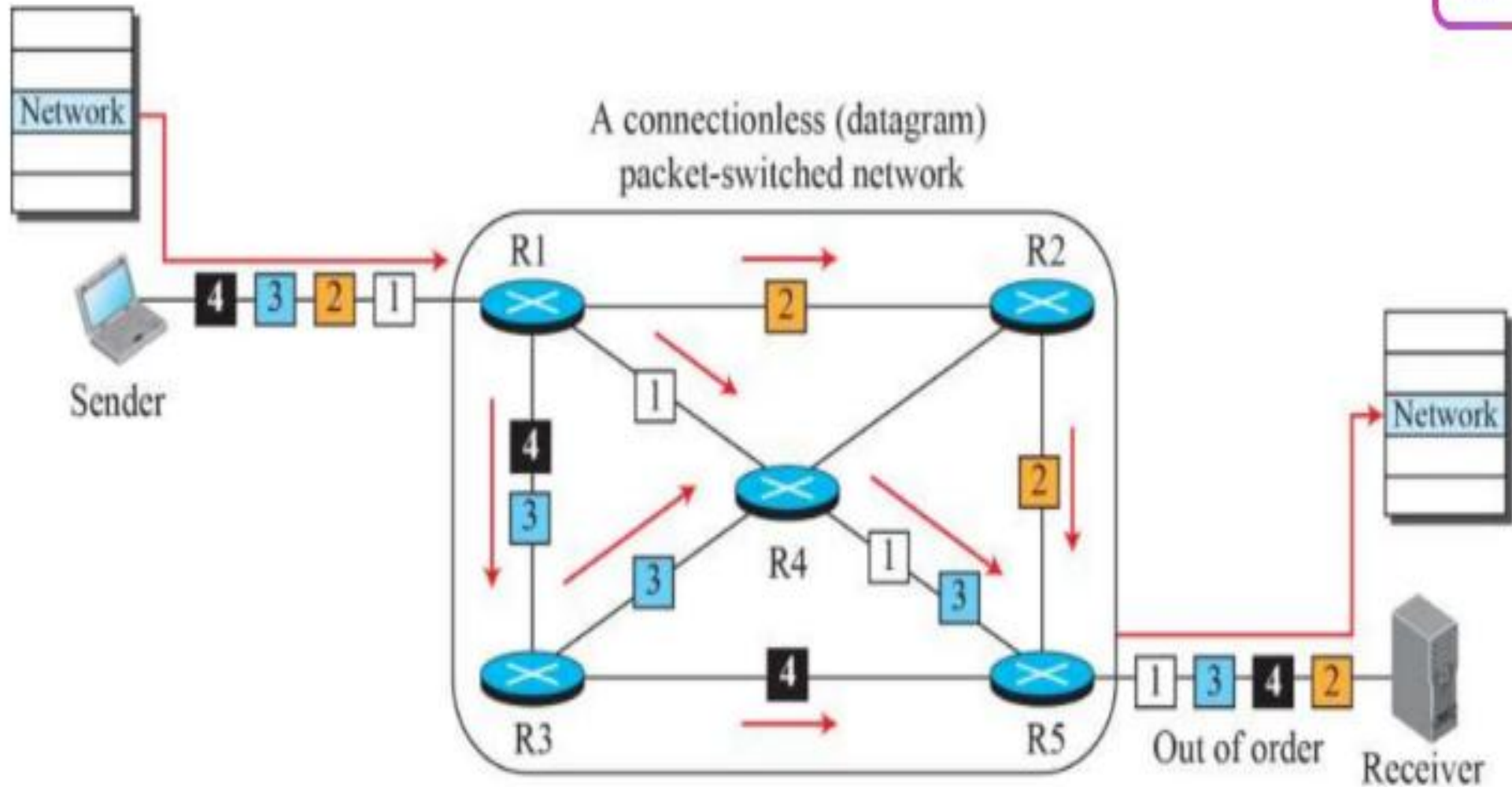


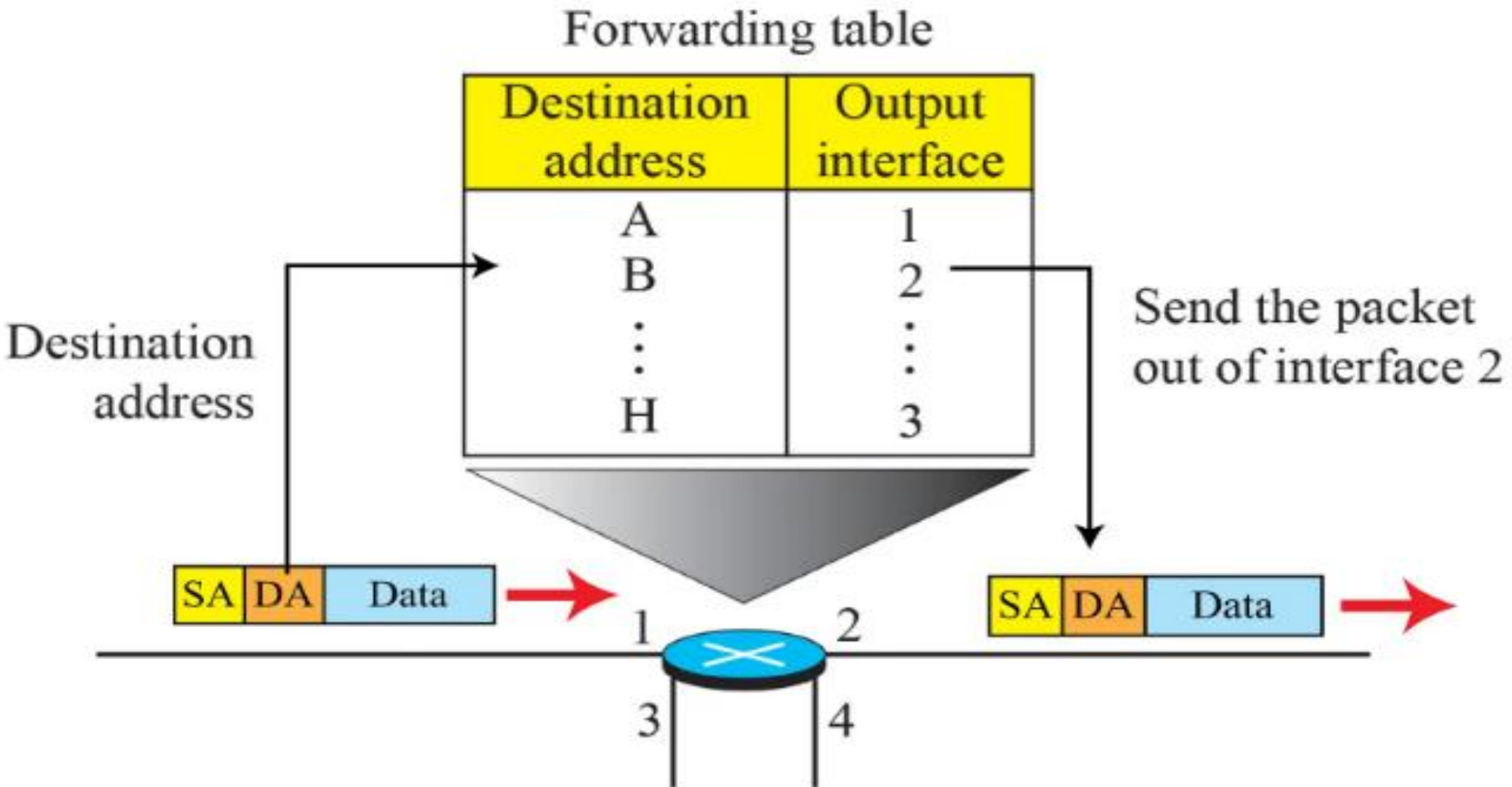
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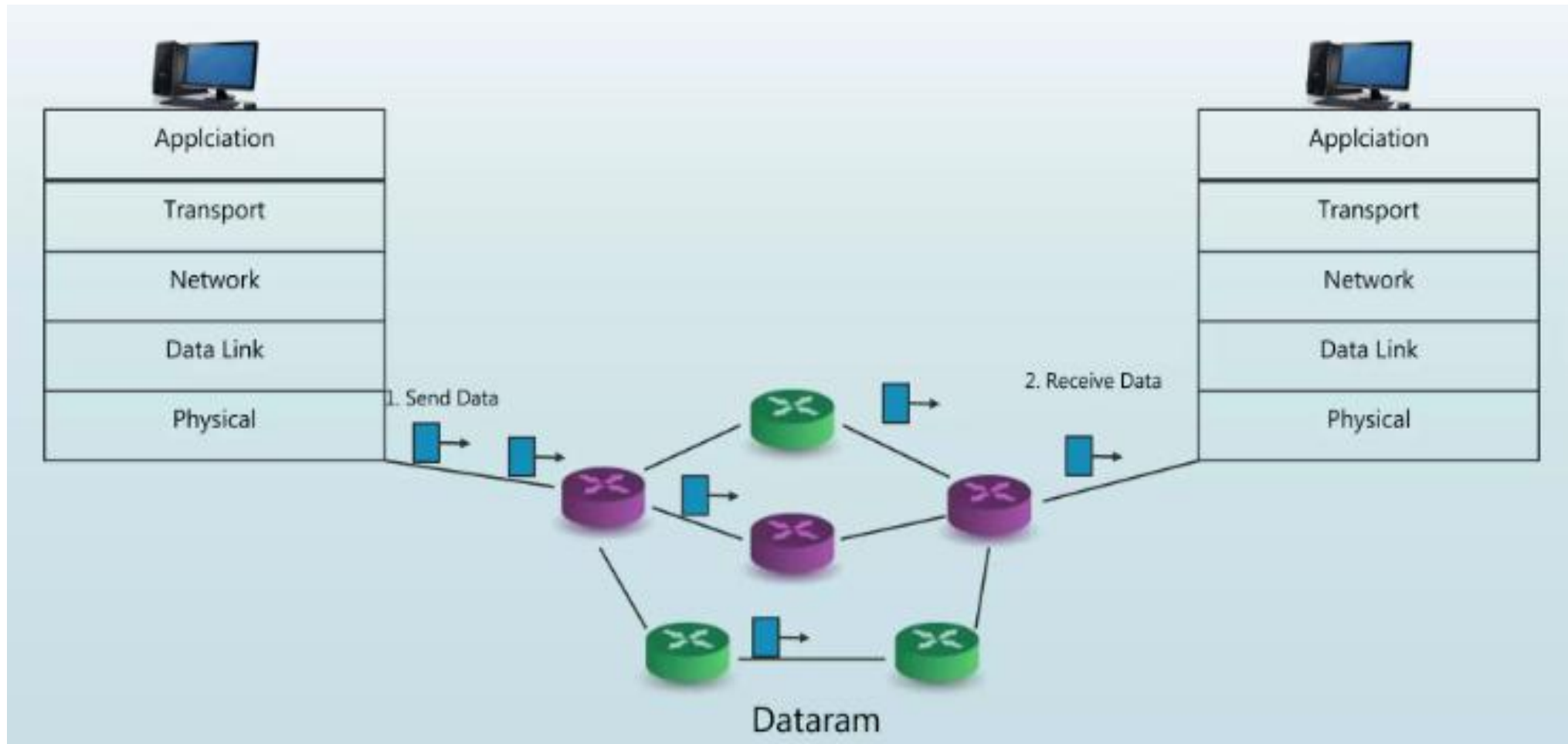
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Datagram





18.2.2 Virtual-Circuit Approach

In a connection-oriented service (also called virtual-circuit approach), there is a relationship between all packets belonging to a message. Before all datagrams in a message can be sent, a virtual connection should be set up to define the path for the datagrams. After connection setup, the datagrams can all follow the same path. In this type of service, not only must the packet contain the source and destination addresses, it must also contain a flow label, a virtual circuit identifier that defines the virtual path the packet should follow.



Virtual Circuit

- An initial setup phase is used to set up a route between the intermediate nodes for all the packets passed during the session between the two end nodes.
- In each intermediate node, an entry is registered in a table to indicate the route for the connection that has been set up.
- Thus, packets passed through this route, can have short headers, containing only a virtual circuit identifier (VCI), and not their destination.
- Each intermediate node passes the packets according to the information that was stored in it, in the setup phase.
- In this way, packets arrive at the destination in the correct sequence, and it is guaranteed that essentially there will not be errors.
- This approach is slower than Circuit Switching, since different virtual circuits may compete over the same resources, and an initial setup phase is needed to initiate the circuit. As in Circuit Switching, if an intermediate node fails, all virtual circuits that pass through it are lost.
- The most common forms of Virtual Circuit networks are X.25 and Frame Relay, which are commonly used for public data networks (PDN)

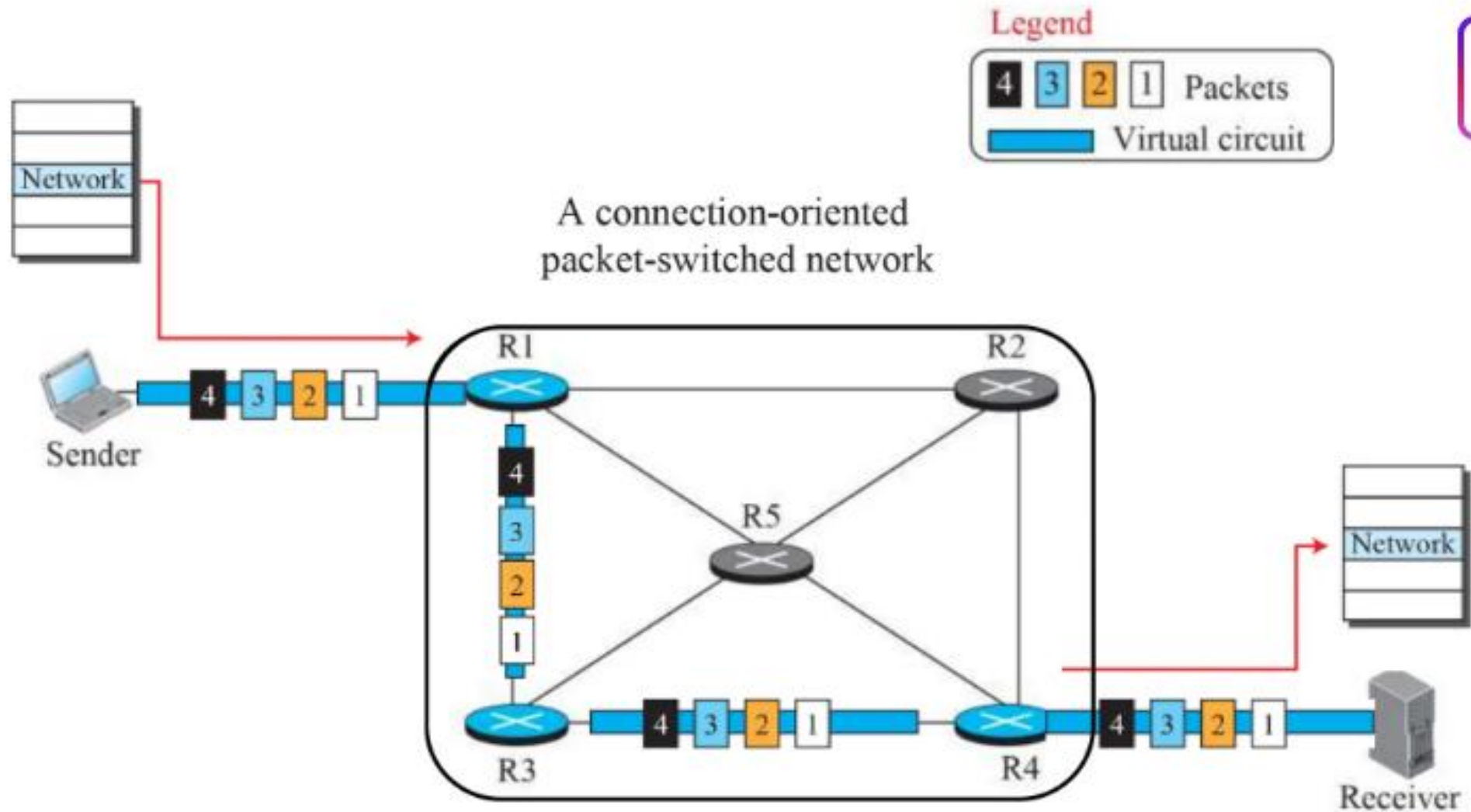


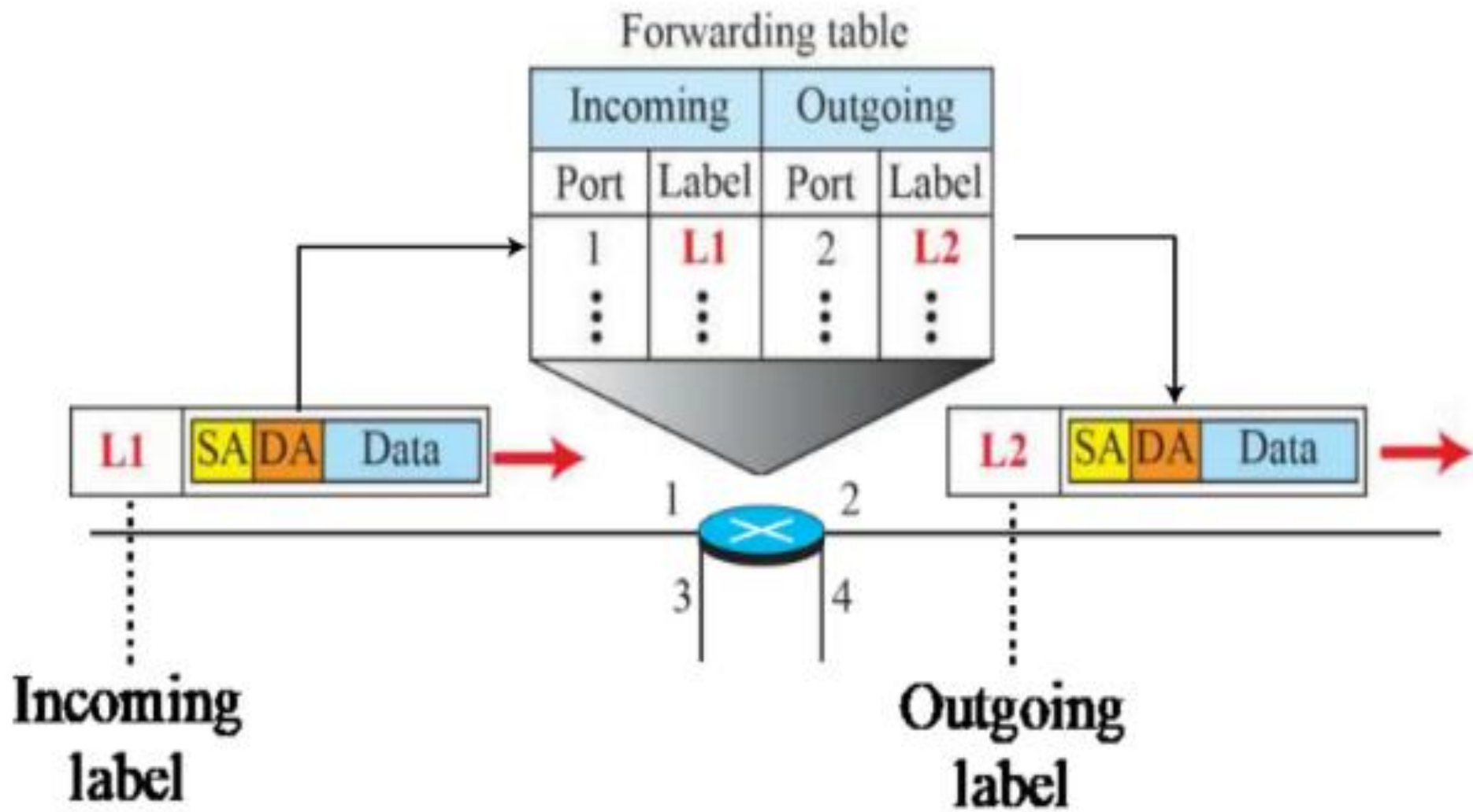
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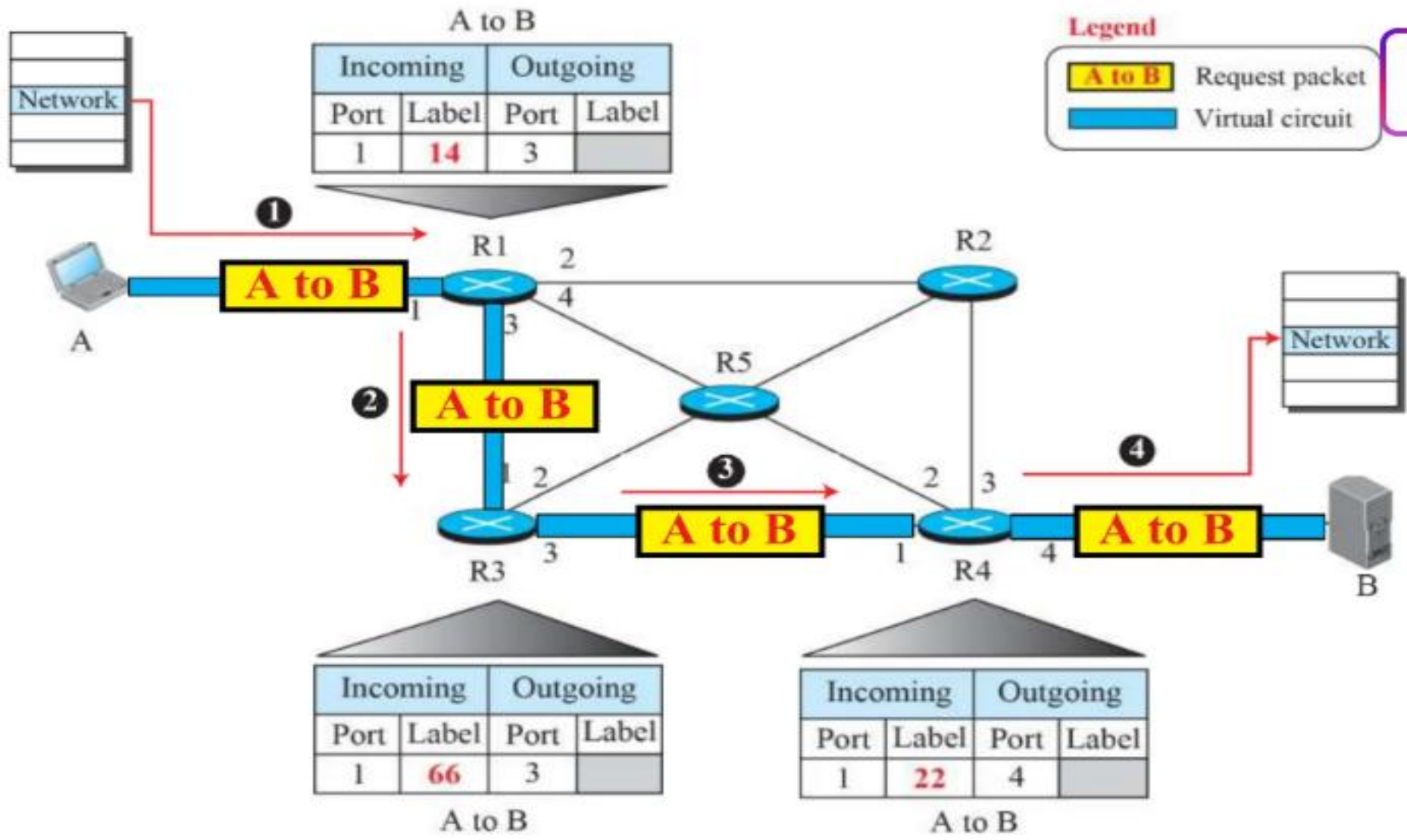
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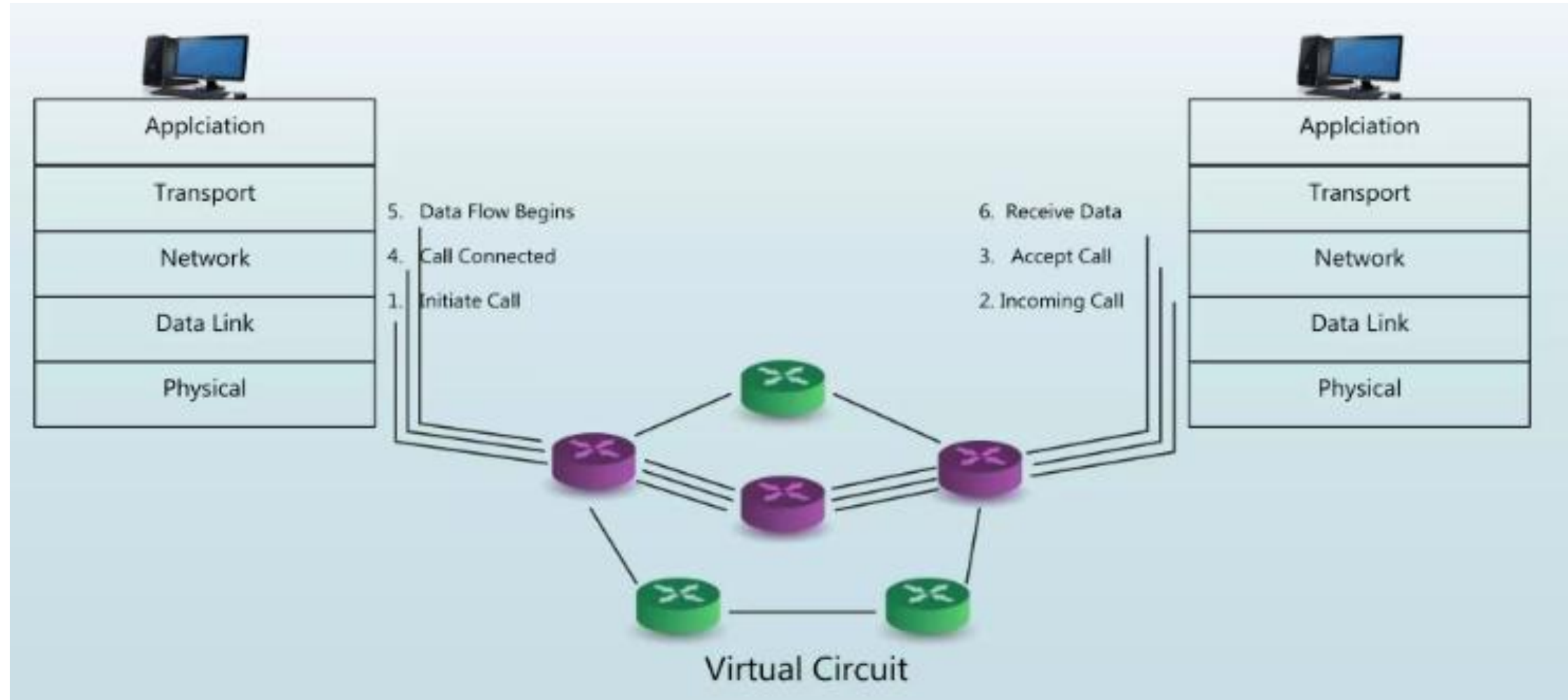
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Virtual Circuit





Datagram VS Virtual Circuit Packet Switching

Datagram Packet Switching	Virtual-circuit Packet Switching
Two packets of the same user pair can travel along different routes.	All packets of the same virtual circuit travel along the same path
The packets can arrive out of sequence	Packet sequencing is guaranteed
Packets contain full Source, Destination addresses	Packets contain short VC Id. (VCI)
Each host occupies routine table entries	Each VC occupies routing table entries
Requires no connection setup	Requires VC setup. First packet has large delay
Also called Connection less	Also called connection oriented
Eg. Internet which uses IP Network protocol	Examples: X.25 and Frame Relay



18.3.1 Delay

All of us expect instantaneous response from a network, but a packet, from its source to its destination, encounters delays. The delays in a network can be divided into four types: transmission delay, propagation delay, processing delay, and queuing delay. Let us first discuss each of these delay types and then show how to calculate a packet delay from the source to the destination..



18.3.2 Throughput

Throughput at any point in a network is defined as the number of bits passing through the point in a second, which is actually the transmission rate of data at that point. In a path from source to destination, a packet may pass through several links (networks), each with a different transmission rate. How, then, can we determine the throughput of the whole path? To see the situation, assume that we have three links, each with a different transmission rate, as shown in Figure 18.10.

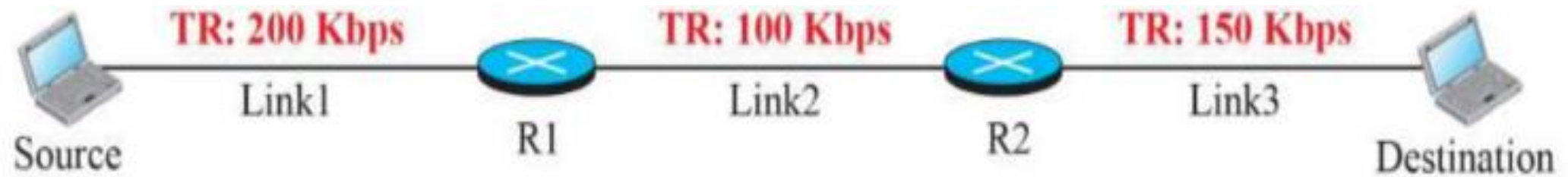


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a. A path through three links



b. Simulation using pipes

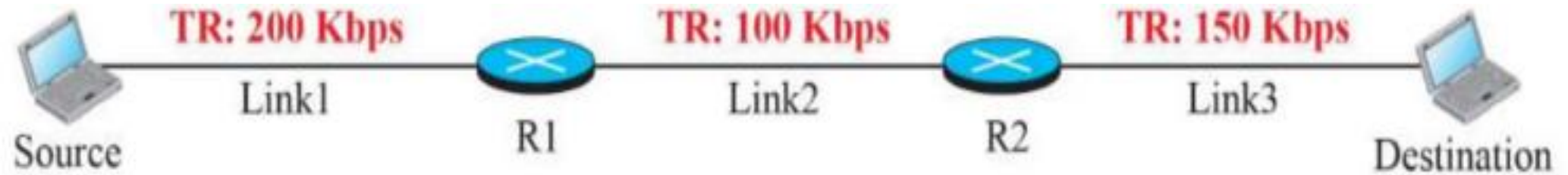


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a. A path through three links



b. Simulation using pipes



18.3.3 Packet Loss

Another issue that severely affects the performance of communication is the number of packets lost during transmission. When a router receives a packet while processing another packet, the received packet needs to be stored in the input buffer waiting for its turn. A router, however, has an input buffer with a limited size. A time may come when the buffer is full and the next packet needs to be dropped. The effect of packet loss on the Internet network layer is that the packet needs to be resent, which in turn may create overflow and cause more packet loss.